

Vibration SENSOR

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Presented here is a sensitive vibration alarm for use as a simple surveillance system for protecting doors and windows. It can also be used as a luggage or locker protector. It beeps and lights a white LED when it detects even a slight vibration. It is compact, battery-operated and can be enclosed in a small box.

The circuit uses a miniature vibration sensor SW18020 P from Gaoxin. It can be used in different ways to sense mechanical vibrations to activate alarms and other surveillance systems in a variety of vibration-detection projects.

The vibration sensor has two electrical contacts that do not touch each other in idle condition. When any movement or vibration occurs, the sensor's contacts close and touch each other. When the movement or vibration stops, the sensor's contacts return back to their original positions, away from each other. The closed contacts during vibration trigger the circuit connected to it. The author's prototype is shown in Fig. 1.

The vibration sensor has a small spring mechanism that makes the contacts touch each other when vibration occurs above a certain threshold level. Two pins coming out of the sensor are insulated by a resistance of more than 10-mega-ohm. During vibration the spring inside the sensor vibrates and makes a momentary short-circuit between the two terminals.

Terminals of the vibration sensor have no polarity but one pin is thick. It is connected to Vcc through a resistor and the thin pin is connected to the circuit to be triggered.

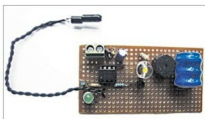


Fig. 1: Author's prototype



Fig. 2: Vibration sensor

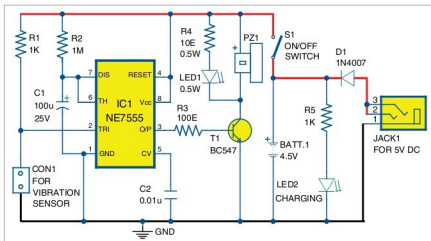


Fig. 3: Circuit diagram of the vibration sensor

The sensor's maximum working voltage is 12V DC but it works even at three volts. When using it in a circuit, it consumes less than 5mA current and offers around 10-mega-ohm contact resistance in open state and less than 5-ohm in contact state. It is highly reliable and its response time is less than 2ms. It works more than 500,000 times without breakdown. The vibration sensor is shown in Fig. 2.

Circuit and working

Circuit diagram of the vibration sensor is shown in Fig. 3. It is built around NE7555 timer (IC1), npn transistor BC547 (T1), piezo buzzer (PZ1) and a few other components.

The circuit is simple. NE7555

PARTS LIST

Semiconductors:

- IC1 - NE7555 timer
- T1 - BC547 npn transistor
- LED1 - White LED, 0.5W
- LED2 - 5mm LED
- D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1, R5 - 1-kilo-ohm
- R2 - 1-mega-ohm
- R3 - 100-ohm
- R4 - 10-ohm, 0.5W

Capacitors:

- C1 - 100 μ F, 25V electrolytic
- C2 - 0.1 μ F ceramic disk

Miscellaneous:

- CON1 - 2-pin connector
- JACK1 - DC power jack
- PZ1 - Piezo buzzer
- S1 - On/off switch
- BATT.1 - 4.5V rechargeable battery
- Vibration sensor SW18020 P
- 5V DC power supply
- adaptor with power plug